

High Temperature Oxide Dispersion Strengthened (ODS) Alloy Mechanical Testing
Statement of Work (SOW)

The National Aeronautics and Space Administration (NASA), specifically Marshall Space Flight Center (MSFC) has a need for mechanical testing with unique NASA-developed alloys with an independent laboratory to validate material properties for design.

The SOW is as follows:

1. The vendor shall receive all samples and inspect samples numbers.
2. The vendor shall track all samples (ie., bag and tag samples) in preparation for machining.
3. The vendor shall machine and perform mechanical testing of all samples according to a matrix provided below.
4. **The vendor shall complete all machining, mechanical testing (fatigue) within 16 weeks of receiving all samples at their site.**
5. The vendor shall be, and maintain, Nadcap and A2LA (ISO/IEC 17025:2017) accredited in the use of ASTM E8/E21 and E606/E466.
6. The vendor shall have experience testing oxide dispersion strengthened (ODS) alloys including GRX-810, ORCAloy-21 and Alloy 754-ODS and compatible data.
7. The tensile specimens shall be prepared for testing and machined per dimensions according to E8/E21. A minimum of (165) samples will be provided fully machined for tensile testing.
8. The vendor shall complete machining and grinding of all low cycle fatigue (LCF) specimens, unless otherwise noted as being tested in the as-built surface (75 total LCF) according to ASTM E606 samples geometry and according to drawings PES-FAT-.25GD-TH-SHORT for elevated temperature specimens and PES-FAT-.25GD-SM-SHORT for room temperature; samples are ~0.55" diameter and 3.25" length. Machining shall be controlled to meet all dimensions.
9. Vendor shall complete low cycle fatigue testing at room and elevated temperatures up to 2100F on (75) specimens according to ASTM E606.
10. Vendor shall complete tensile testing of (65) room temperature samples according to ASTM E8 and tensile testing of (100) samples at temperatures up to 2100F according to ASTM E21.
11. The vendor shall complete machining and grinding of all high cycle fatigue (HCF) specimens, unless otherwise noted as being tested in the as-built surface (75 total HCF) according to ASTM E606 samples geometry and according to proprietary drawings; samples are a minimum of ~0.6" diameter and ~4" length. Machining shall be controlled to meet all dimensions.
12. Vendor shall complete high cycle fatigue load control testing at temperatures up to 2000F required (75 total HCF) on specimens according to ASTM E466.
13. All machining and testing shall be performed onsite of the specified company holding the contract. It is not permissible to subcontract any of the related testing. Any testing not performed onsite at the company holding the contract must be approved by NASA.
14. Test overruns for LCF and HCF should be reported to NASA for early awareness to allow

adjustments as necessary.

15. Vendor shall provide updates on fatigue testing to allow adjustments to conditions if necessary.

16. NASA will provide detailed test matrices and discuss as necessary.

Shipping

1. Contractor shall ship all deliverables and returned samples to:

Marshall Space Flight
Center Central Receiving,
Bldg 4631
Huntsville, AL 35812

Deliverables

1. Final test report with all summary data.
2. Digital data as requested (such as csv or excel data for stress/strain and hysteresis from each sample along with the report.).
3. Raw data including hysteresis curves as requested.